Qiancheng (Sam) Sun

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EDUCATION

Purdue University | West Lafayette, IN *2022-2027*
Bachelor of Science, Mechanical Engineering GPA: 3.86 / 4.00

EXPERIENCE

Tesla | Fremont, CA *Aug 2026 – Jan 2027*
Mechanical Design Engineering Intern – Cybercab Interior

- Applied **FEA** with multibody dynamics and nonlinear structural analysis, alongside a **capstan**-based torque simulator, to correlate Cybercab armrest **detent** profiles with user effort, accelerating development by weeks.
- Designed and prototyped next-gen Cybercab mechanisms, optimizing detents to reduce user effort by **55**% and increase holding force by **15**%, and developing push-push mechanisms for tight packaging constraints.
- Owned mechanical design iterations across multiple **injection molded** components, driving improvements in part consolidation and installation efficiency.
- Evaluated foam and PSA combinations, reducing high-temp compression set from **20 mm to 1 mm** while maintaining adhesive performance

Apple | Cupertino, CA *May 2026 – Aug 2026*
Product Design EPM Intern – Macbook

- Drove DVT and production ramp of a Class A **injection-molded** component; led parallel qualification paths and convergence plan for PSA and press-force solutions, increasing PSA wetout area by **60%**.
- Developed an AI-assisted tooling should-cost model for new ID programs, established benchmark rates through cross-functional and supplier discussions to support data-driven negotiations; projected **10%** tooling savings and **70%** less negotiation time

Formlabs | Somerville, MA *Jun 2025 – Aug 2025*
Manufacturing Design Engineering Intern – SLA & SLS Printers

- Owned the full **electromechanical** design of a scalable automated solution that accurately measures linear rail drag force and straightness for all current and future SLA printers with a repeatability of **0.3N** and **5 microns**. Reduced rework frequency by **~20%**
- Performed hands-on prototype assembly and testing for a new large-format **SLS** product line; worked closely with design teams to implement **DFA** improvements, reducing assembly time by ~33% on targeted subassemblies
- Conducted **Gauge R&R** on a planarity jig and used **CMM** to root cause errors. Analyzed results in **JMP** to validate jig performance

Tesla | Austin, TX *Aug 2024 – May 2025*
Manufacturing Engineering Intern – 4680 Cell

- Led the standardization & rollout of a precision air jet system, replacing flawed hardware for a critical step of **4680** winding process
  - Performed **DOE** to identify and eliminate unnecessary components on gimbal; reduce wear and increased the process window by 100+%; conducted **tolerance stack analysis** to ensure consistent performance across manufacturing variation
  - Spearheaded first-principles investigation into hi-pot defects caused by Gen 2 air jets; used **CFD** and high-speed footage to isolate root cause and deploy solution, restoring **~1% yield** fleet-wide
  - Identified and redesigned critical components to resolve 12+ **DFA**-related failures and prevent recurrence
- Collaborated with **OEM** to design custom one-way rollers with fiber optic sensors for missed roller detection, reducing scrap by 10,000+ cells/month; iterated designs through supplier **DFM** reviews to deliver a low-cost, manufacturable solution
- Built custom **SQL** queries & **Tableau** dashboards to monitor downstream impact of design changes, supporting data-driven decisions

Panasonic Energy of North America (PENA) | Sparks, NV *May 2024 – Aug 2024*
Equipment Engineering Intern – 2170 Cell

- Designed and procured a 3-axis adjustable vision-inspection bracket using **DFMA**, **GD&T**, and welding callouts, delivering a production-ready design 32% under budget for high-speed 2170 cell winders

Projects

Accessible 3D Printed Prosthetic Hand | *Formlabs Hackathon* *2022*

- Designed a <$200 prosthetic hand using single servo and 3D printed gears reduce cost and simplify assembly for broader accessibility
- Engineered compliant fingers with flexible resin and elastic couplings to enable adaptive gripping behavior
  Designed a modular arm interface with integrated electronics housing and adjustable fittings for diverse limb geometries.

AI Powered Smart Trash-Can | *MakeMIT 2023 | MIT Hardware Hackathon* *2023*

- Built and presented prototype trash can within 24 hours; Second place MakeMIT award, 1517 Medici Grant Recipient
- Auto-opening lid, Computer Vision powered object recognition, Automatic Trash sorting

SKILLS

CAE: Catia V6, 3DX, SolidWorks, OnShape, ANSYS, NX, COMSOL, FEA, GD&T | **Fluent Languages:** Mandarin & English
Mechanical: Design for Manufacturing/Assembly (CNC, Sheet Metal, Injection Molding, 3D Printing), Root Cause Analysis, FMEA
Software/Electrical: Python, Matlab, C++, Linux, Arduino, Raspberry Pi, Soldering, SQL, JMP, Tableau, JIRA, Confluence